MATRICA

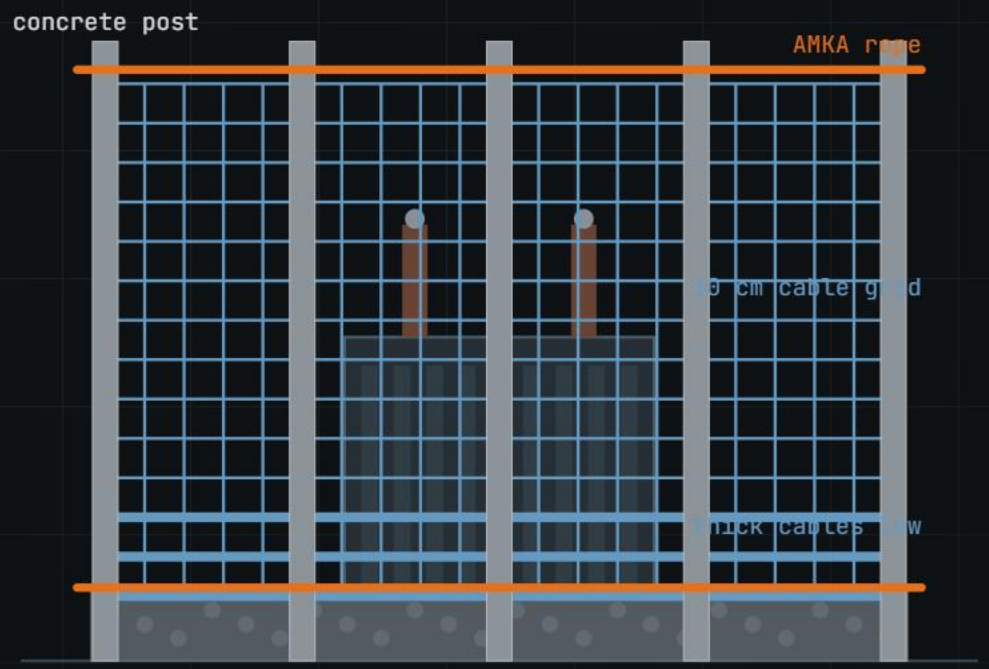
Grid waste becomes grid shield.

We take the cables and concrete poles grid companies throw away, and build drone screens around small, critical grid equipment.

WE TAKE Cable offcuts + Concrete poles → Drone shield

Climathon 2026 · RTU

Challenge: AS "Sadales tīkls"



crushed-concrete wall · 2 m standoff · earthed

THE PROBLEM

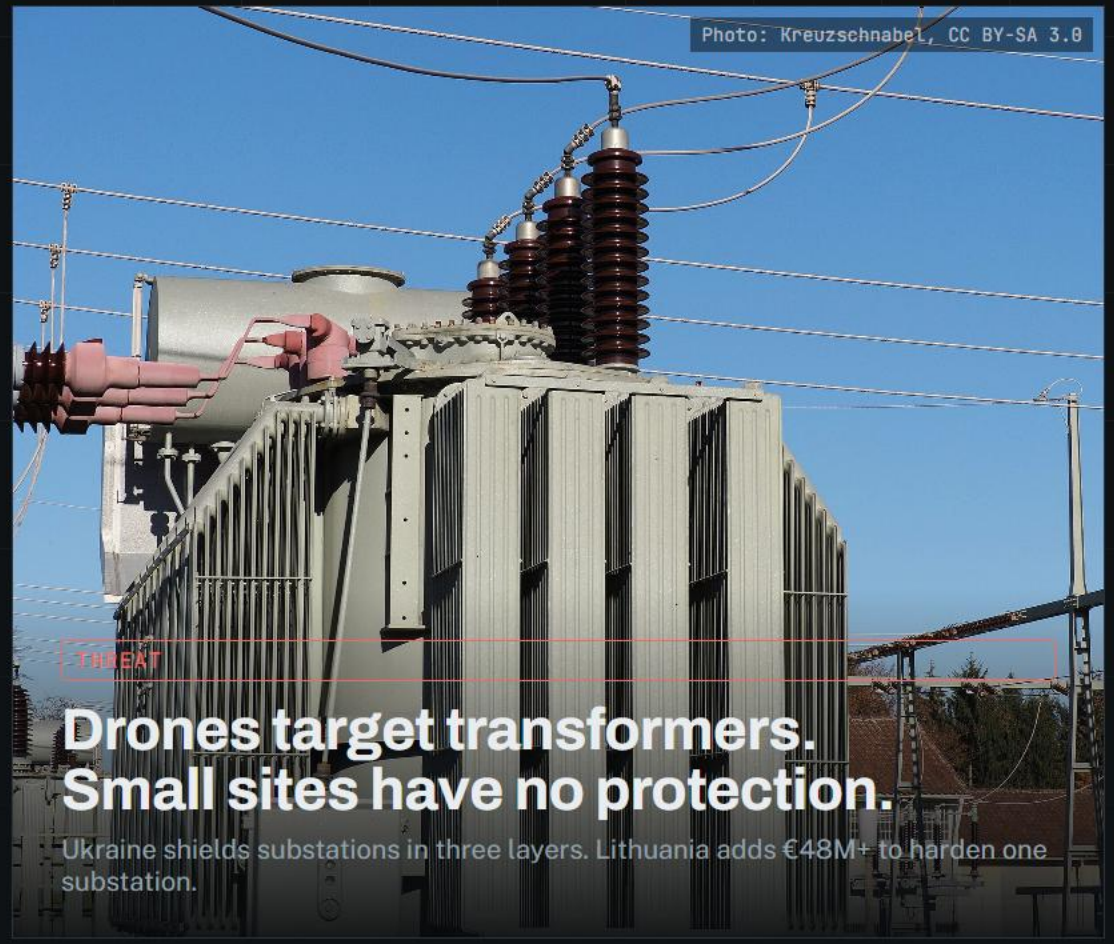
Two problems, one place.



WASTE

Offcuts too short to reuse. Concrete poles that crumble when removed.

Sadales tīkls challenge brief: no reuse today for broken concrete poles.



THREAT

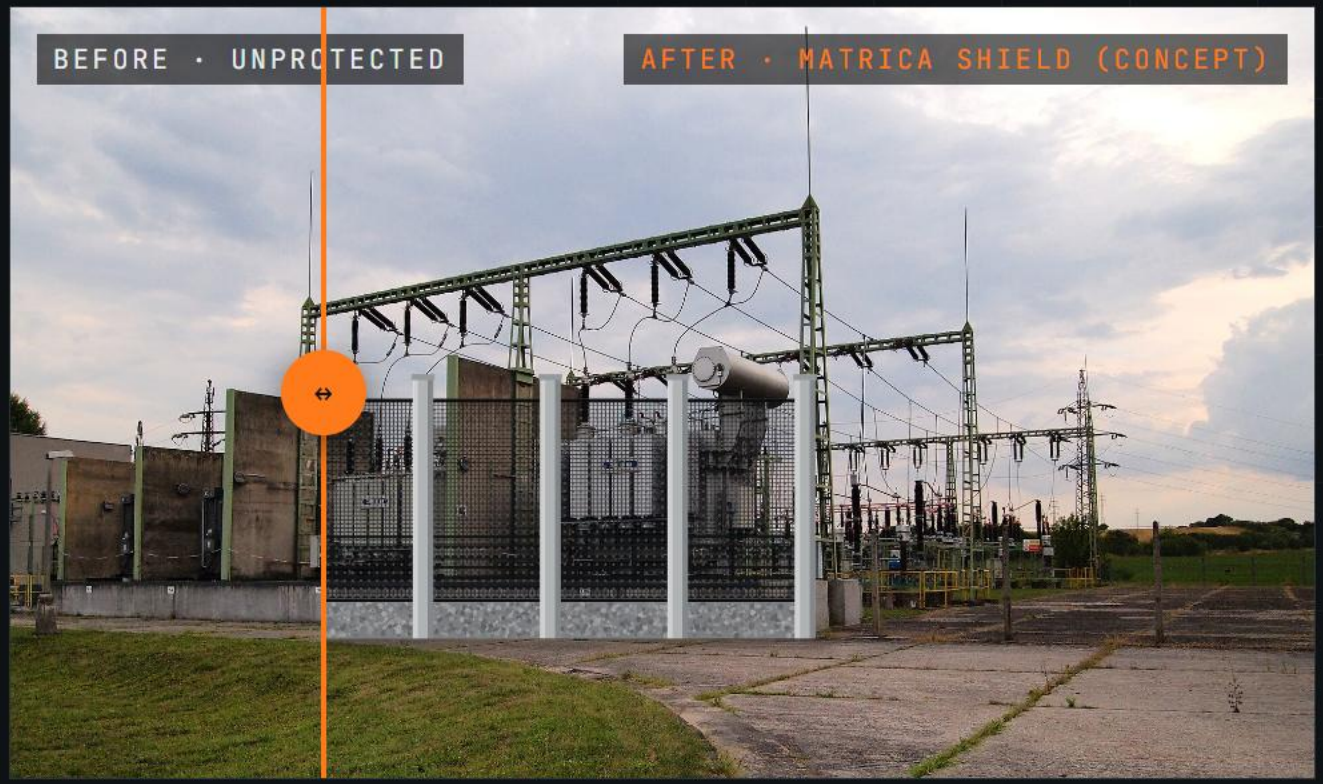
Drones target transformers. Small sites have no protection.

Ukraine shields substations in three layers. Lithuania adds €48M+ to harden one substation.

THE SOLUTION

A cable lattice on concrete posts.

WE TAKE Cable offcuts + Concrete poles → Drone shield



Drag the handle. Same real substation: before, and after with our shield (concept visualisation): concrete posts, a cable grid and a crushed-concrete wall. Photo: Tiia Monto, CC BY-SA 4.0; shield drawn by Matrica.

→ Cable offcuts

Whole, insulation on, laid straight in a 10 cm grid and tied at every crossing.

→ Concrete poles

Intact ones become posts 2 m out. Broken ones are crushed into a low wall.

→ Overhead AMKA cable

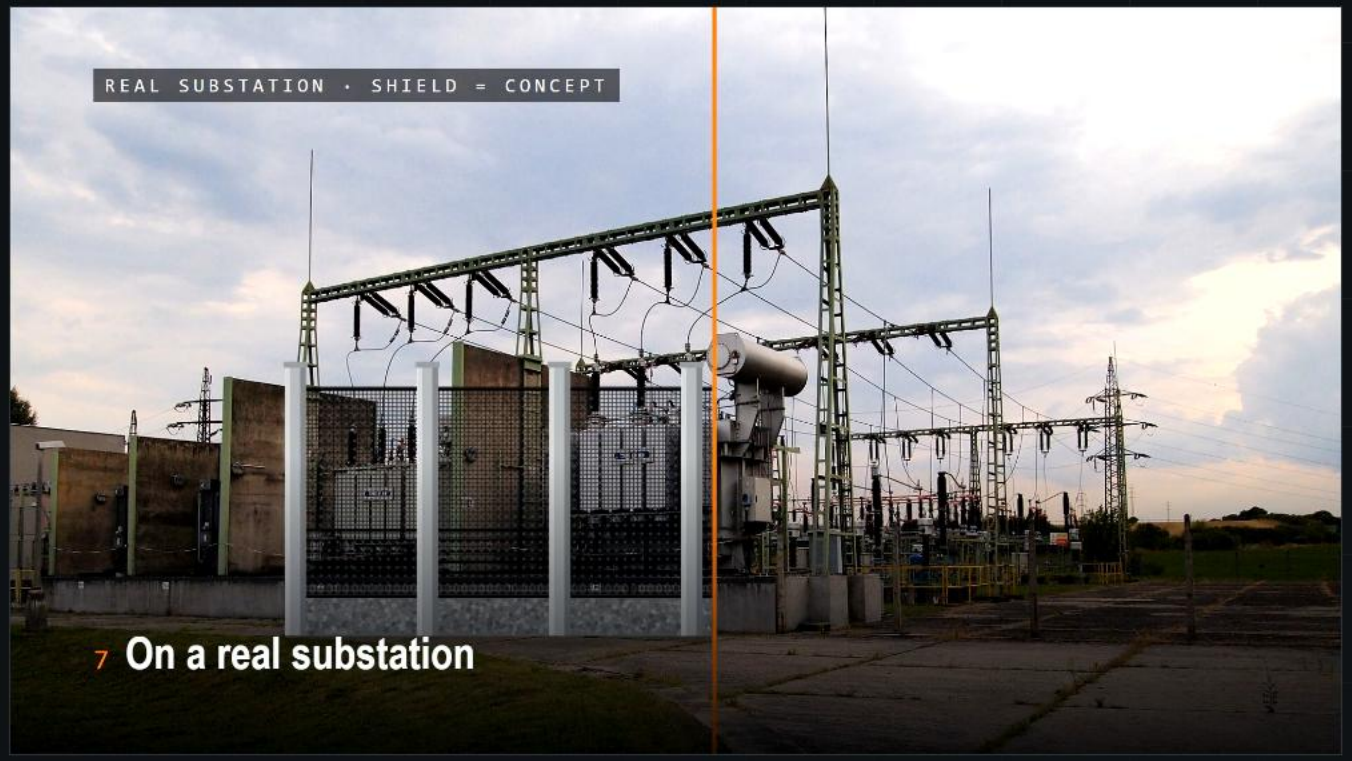
Used whole as the top and bottom rope.

No stripping. No bending. Any length, any size.

Built for Baltic weather: wind, ice, snow, rain and sun.

HOW IT WORKS

From bin to shield in six steps.



1 **Collect**
Crews throw offcuts in one bin. No new work.

2 **Check**
Reject only metal that melted or crumbles.

3 **Sort**
Thin → grid. Thick → low slats. AMKA → rope.

4 **Set posts**
Concrete posts 2 m out, crushed-concrete wall.

5 **Build**
Straight cables, 10 cm grid, tied at crossings.

6 **Earth & inspect**
Outside the 110 kV zone. Yearly check.

One kit per transformer · 3-person crew · about 4 days ESTIMATE

WHERE

Small, critical objects across the Baltics.



ESTONIA · SUPPLIERS & BUYERS

Elektrilevi | Telia

LATVIA · SUPPLIERS & BUYERS

ST | AST | elektrum group
conexus | GASO | LVRTC
Tet | LMT | Tele2
Bite | LDZ

LITHUANIA · SUPPLIERS & BUYERS

eso | Litgrid | ignitis

TOP TARGETS · SCORED /60

110/20 kV substation transformers 46
AST / Sadales tīkls

20/0.4 kV kiosks feeding hospitals and water 45
Sadales tīkls

Grid battery storage containers 44
AST

Hover a logo to see what the company would supply or buy. Red rings: Latgale pilot region.

Target suppliers and customers. No agreements yet. Logos belong to their owners. Positions approximate.

ECONOMICS

One kit. Clear numbers. ESTIMATES

€10–14k
our cost per transformer kit

€18–25k
price per kit

~10%
yearly inspection contract

~€0
material cost: it's their waste

TRY IT · MARKET MODEL

Price per kit: €20k

Transformers protected: 275

Latvia ≈ 250–300 transformers. Baltics ≈ 3× that. Cost assumed €12k per kit.

€5.5M
kit revenue

€2.2M
gross margin

€550k
service per year

↓
disposal cost for the grid company

PRACTICAL

Safe, weatherproof, built with what crews know.

Safe

Earthed lattice

Outside the 110 kV clearance zone

Built under the site's work permit

Weatherproof

Open 10 cm grid: low wind load, snow falls through

Concrete posts carry ice on the cables

Outdoor cable sheaths: UV- and rain-proof, no rust

Checked after every storm

Honest limits

Stops FPV and small drones, debris

Does not stop Shahed-class drones or missiles

Version 1 covers the sides

Pilot

2–5 transformers in Latgale

With Sadales tīkls

Measure cost, time, tonnes reused and winter performance

TARGET SUPPLIERS, CUSTOMERS AND PARTNERS

The companies that would want this.

MATERIAL SUPPLIERS AND BUYERS

ST	AST	elektrum group	conexus BALTIC GRID	GASO	LVRTC
Tet	LMT	Tele2	Bite	LDz	Elektrilevi
elering	Telia	eso	Litgrid	ignitis grupė	

DELIVERY PARTNERS

JAUDA	ELTEL	VINCI ENERGIES	Eco Baltia	NKT	prysmian	SAF Tehnika	Origin Robotics

EUROPEAN FUNDS FOR SCALE-UP

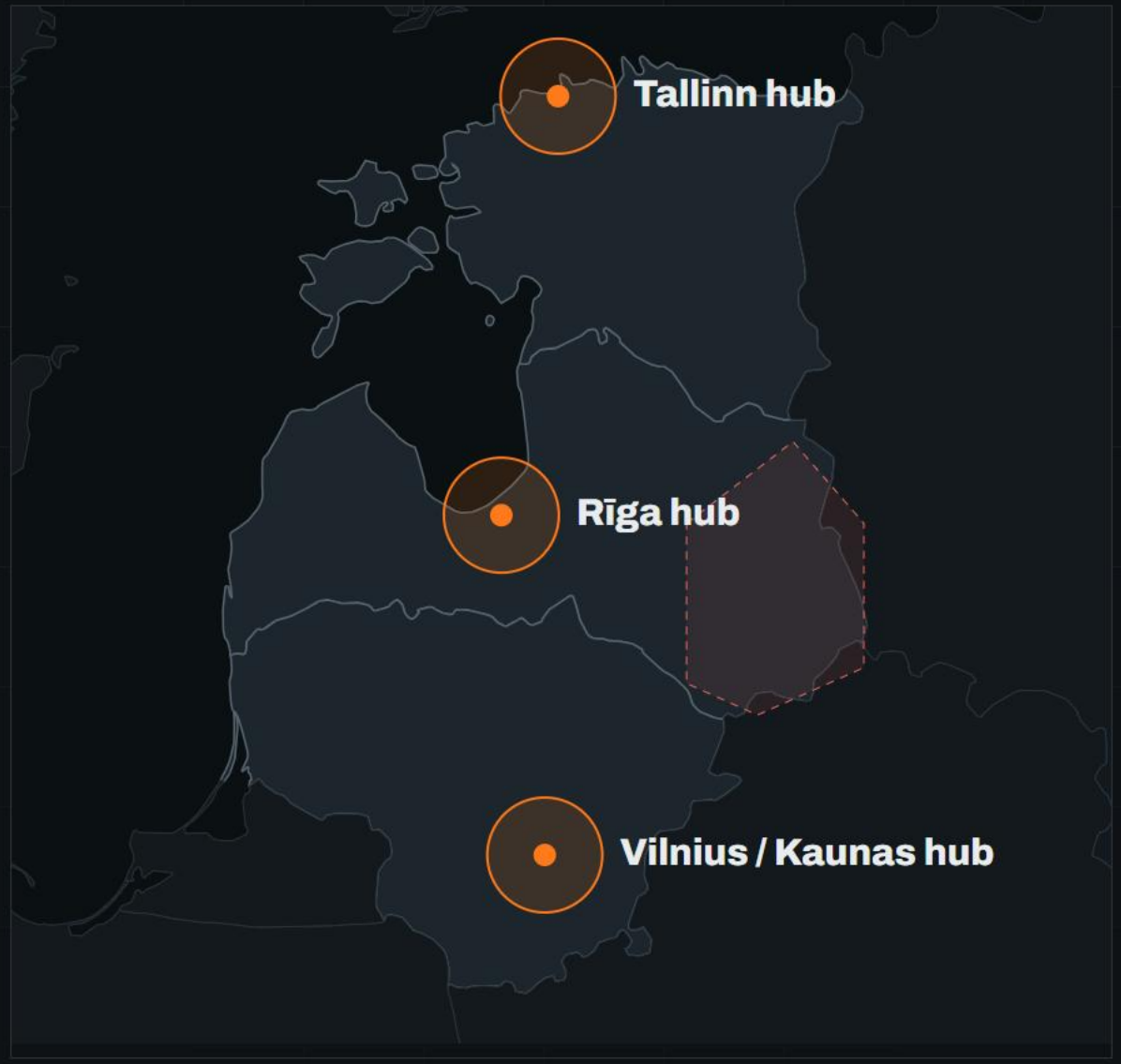
- **Latvian dual-use programme** LIAA / ALTUM, EU co-financed
- **LIFE** Circular economy, material reuse
- **Horizon Europe Cluster 3** Critical infrastructure protection
- **EIC Accelerator** Grant + equity to scale
- **Interreg Baltic Sea Region** Three-country rollout

No agreements yet. Logos belong to their owners and are shown to name target companies.

THE FUTURE

A Baltic take-back network.

Three hubs. Every grid, telecom and renewables company as a supplier. Each country's waste protects its own grid.



OUR ASK

A pilot with Sadales tīkls on 2–5 transformers.

APPENDIX

Scored target objects (0–10 each)

Object	Owner	Crit	Fit	Exp	Vol	Safe	Pay	/60
110/20 kV substation transformers	AST / Sadales tīkls	9	8	7	8	5	9	46
20/0.4 kV kiosks feeding hospitals and water	Sadales tīkls	6	10	5	9	8	7	45
Grid battery storage containers	AST	9	9	8	2	7	9	44
Gas regulating and metering stations	Conexus / Gaso	7	9	6	7	8	7	44
Drinking-water pumping stations	Municipal utilities	8	9	5	8	9	5	44
Mobile base station cabinets	LMT / Tet / Tele2 / Bite	7	8	6	10	8	5	44
Broadcast transmitter sites	LVRTC	8	7	6	4	7	7	39
Railway traction substations	LDz	7	8	6	5	6	6	38
Small district-heating boiler houses	Municipal	7	6	5	7	8	5	38
Data-centre cooling and generator yards	Tet and others	7	8	4	4	8	7	38
Synchronous condensers	AST	9	5	6	2	6	9	37
Airport navigation aids	LGS	8	7	4	3	8	7	37

Assumptions behind the economics

- Kit for one 110/20 kV transformer: ~14 concrete posts, ~180 m^2 of cable lattice, ~120 m AMKA rope, ~10 m^3 crushed-concrete wall.
- Cost €10–14k: baskets and earthing €2–3k, cable preparation €2–3k, installation 3 people × 4 days €4–5k, engineering and transport €2–3k.
- Price €18–25k and a yearly service at ~10% are hypotheses to validate with Sadales tīkls.
- ~250–300 transformers in Latvia (most 110 kV substations have two). To confirm with AST and Sadales tīkls.

Sources

- Sadales tīkls challenge brief and stock list
  Climathon 2026 RTU materials (this repo)
- Ukraine: three levels of energy protection
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- EIB €86M to Ukrenergo for anti-drone protection
  https://en.interfax.com.ua/news/general/1036088.html
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  https://eutoday.net/lithuania-drone-proof-harmony-link-eu-grid-security/
- AST batteries in Rēzekne and Tume
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- Anti-drone net mesh sizes and twine ratings
  https://ahcebrian.com/en/blog/passive-anti-drone-nets-explained/
- EU CER Directive, Latvian transposition
  https://www.iem.gov.lv/en/article/government-strengthens-critical-infrastructure-resilience-and-national-security
- Baltic defence and dual-use investors
  https://www.vestbee.com/insights/articles/europes-defence-and-dual-use-investors
- Full research with all sources
  research/grid-shield.html

Image and data credits

- **Cable scrap pile:** GeorgHH, CC BY-SA 4.0, Wikimedia Commons: Elektrokabelschrott 0109.jpg
- **Substation before/after:** Tiia Monto, CC BY-SA 4.0, Wikimedia Commons: Milovice - transformer station.jpg. The after image adds a shield drawn by Matrica (concept) and is shared under CC BY-SA 4.0.
- **110/20 kV transformer:** Kreuzschnabel, CC BY-SA 3.0, Wikimedia Commons: Transformer 110 kV 20 kV.jpg
- **Country outlines:** mledoze/countries (GitHub), ODbL, from Natural Earth
- **Company logos:** Wikimedia Commons and the companies' own websites. Logos belong to their owners.
- **Explainer video:** Matrica, made with Remotion (github.com/remotion-dev/remotion)